Operating PostgreSQL as a Data Source for Analytics Pipelines

data egret

Your remote PostgreSQL DBA team

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SECURING YOUR DATABASE AVAILABILITY, SO THAT YOUR TEAM CAN FOCUS ON NEW FEATURE DEVELOPMENT.

- Migrations
- DB audit
- Performance optimisation
- Backup & restore
- Architectural review
- Advising Data Science teams
- Developer training

on premise & cloud

EXPERTISE

Senior DBA with **10+ years** of PostgreSQL administration **experience**

DEVELOPMENT

Involved in **new feature and extension development**

TAILORED APPROACH

Felxible approach and **dedicated team** focused on success of your project

COMMUNITY

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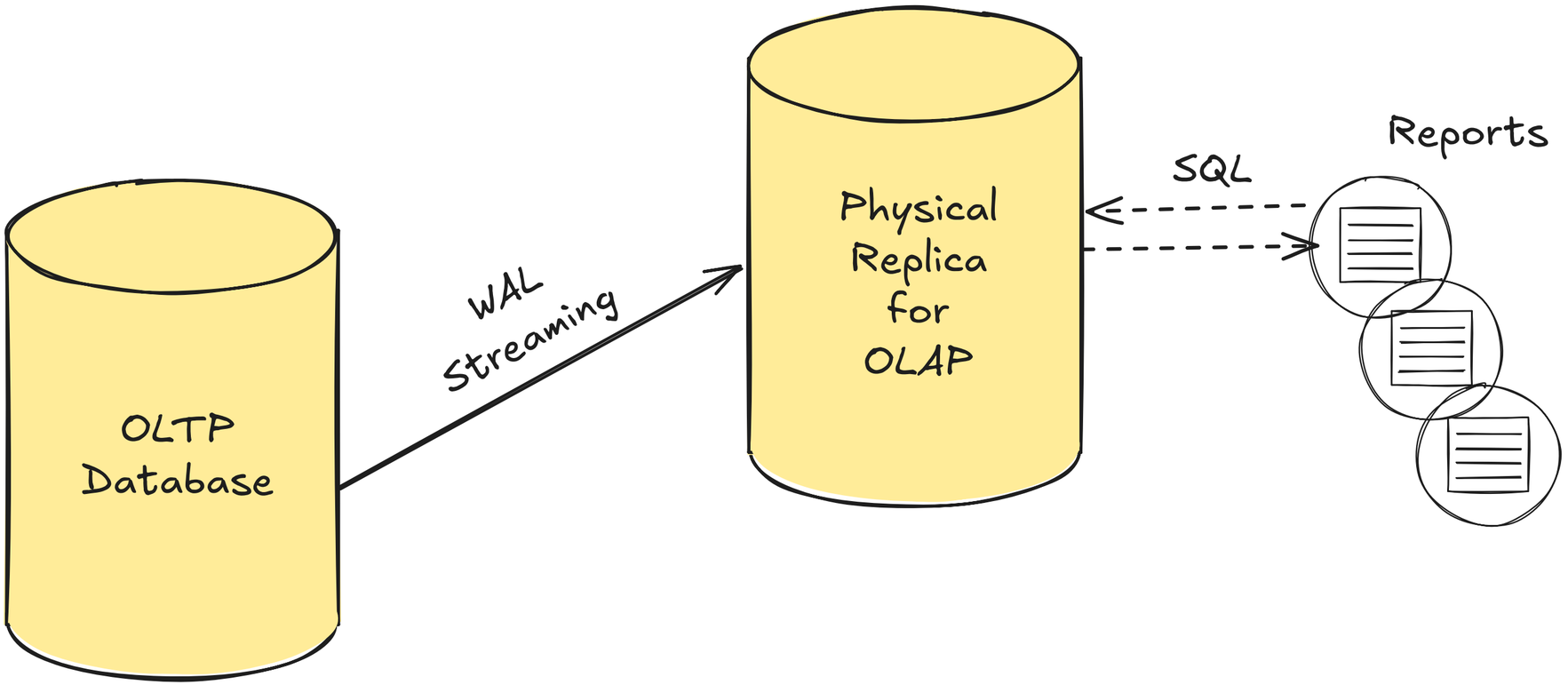
DWH, OLAP, Reporting

Analytics as it used to be

PostgreSQL does it well



Physical Replication (Good):

- Physical Replication is easy to operate
- We have a physical copy of production OLTP database
- We can take full advantage of relational databases to create our reports
- Data scientists can’t ruin your production environment*
- Good performance**

Physical Replication (Bad):

- We have a physical copy of production OLTP database
- …and it is read only
- Lack of useful features on the replica (materialized views, temporary tables…)
- OLTP and OLAP workloads are best friends
- The nature of data and workload can affect performance
- Data scientists can ruin everything!
- Are reports the only analytics we want to have in 2025?

Logical Replication

Solves the problem slightly better

Logical Replication (Good):

- We can have a different schema, a more OLAP-optimized one
- Our DWH can be a subscriber for different sources
- We can retain data which is removed from primary
- No issues during recovery

Logical Replication (Bad):

- We end up with a big RDBMS (which is not always bad, but…)
- DDL (and sometimes DML)
- Failover (especially before PostgreSQL 17)
- Every team *must* be aware of Logical Replication

Operating Logical Replication

Is a topic for a separate talk or rather a training

Christophe made a good one

Before we move onto discussing performance

let’s consider something else

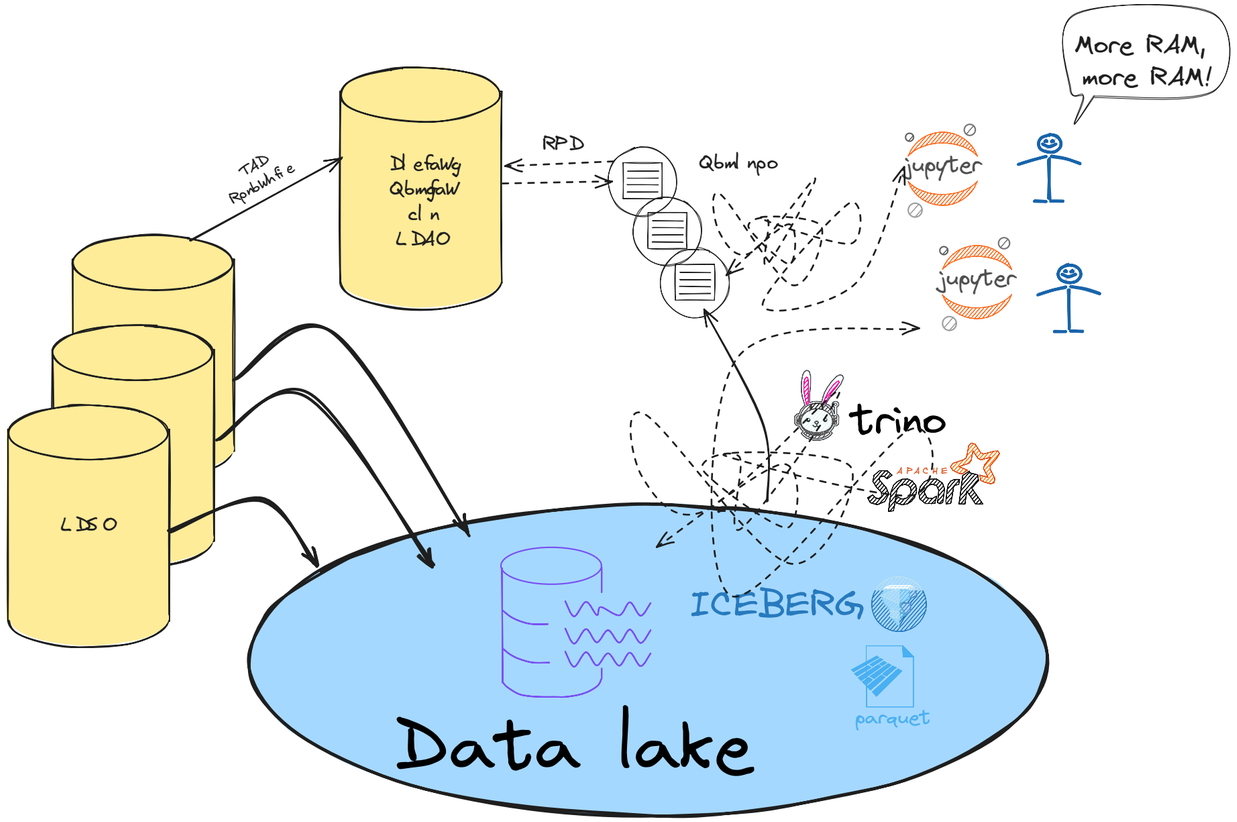
We end up with a big RDBMS

and to figure out if it is good or bad, lets take a step back

What is modern Data Analytics?

- Reports are still there
  - Good old PostgreSQL still can SQL better than many others
- Machine learning
  - Brave new world of Data Lakes and Lakehouses
  - Completely different lifecycle of data
  - Sometimes real-time and mission-critical (anti-fraud, recommendations)
  - Businesses are learning how to earn their money
  - Engineers are learning DataOps and FinOps jobs

Brave new world



ML infrastructure requirements

- Data delivery from operational PostgreSQL
- ML-specific workflows: feature management, DAG storage, enrichment, ingestion, snapshots
- Cost efficiency
- Reliability
- Regulations: for example GDPR and mandatory data retention

What can we use as a Data Lake?

All-in-one solution

Amazon Redshift, GCP Big Query, Snowflake

- Very attractive
- Data delivery is still an open question
- Could easily become *extremely* expensive and barely manageable

PostgreSQL

- Easy to manage
- Data delivery is less of a problem
- ML-specific workflows? That’s a big ‘if’

PostgreSQL: ML-specific workflow

- pgvector allows to store features in PostgreSQL
  - but its job is to provide quick vector search
  - some enrichment capabilities
  - despite all improvements, indexes are not particularly small
- Enrichment, ingestion and generally ETL could easily lead to severe performance issues
- Do we really need a DBMS for a data which fits into S3/Iceberg/Parquet?

Better Data Lake

- is usually a combination of more specific technologies (Iceberg, Spark, Hive, Trino, Clickhouse…)
- We still need to deliver operational data

Change Data Capture and PostgreSQL

- ctid-based, Logical-Replication-based
- Could include some enrichment
- There are low code and efficient solutions

Before you start

- PostgreSQL instance must be well tuned
  - Especially WAL and autovacuum
  - Here is a great guide to PostgreSQL maintenance
- Plan carefully what to capture with CDC
- Workbooks:
  - What to do if CDC is stuck?
  - How to resume/resync after an incident
  - Upgrades and failover/switchover

Airbyte

- A typical example of easy low code CDC solution for PostgreSQL to many destinations
- PostgreSQL connector is based on Debezium and we can’t configure it a lot because of no-code approach
- The ‘let’s convert everything to JSON’ approach is not particularly efficient
- Not all destinations are available in the open-source version

PeerDB and ClickPipes

- PeerDB is open under the Elastic License; ClickPipes is cloud-first
- A combination of ctid and logical decoding can significantly speed up the initial load

Debezium

- Most mature CDC tool
- Can be used as standalone server, but it is not the best idea
  - Scalability is a problem
  - Much better tools like Kafka Connect/Kafka Streams and Apache Flink use embedded Debezium as a library

Kafka Connect

- Brings us the full power of Kafka
- We need to know how to operate Kafka
- It is easy until it is not (or until it is just CDC, not stream processing)
- Kafka Streams can add stream processing to CDC, which is good for ML infrastructure
  - But it is primarily a framework for writing your own software (and operating it!)

Apache Flink

- Client/Server architecture but you still need to write your jobs in Java
- Scales well
- A lot of options for performance tuning
- Extremely performant and handles backpressure
- Swiss Army knife in terms of stream processing, so enrichment could be very complex and still fast

Questions?

Interested in diving deeper? Download our CDC Whitepaper

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